Precept 5 - Solutions

Midterm 1 2021 - Q1 (Yogurt)

The dining hall serves (and you eat) one flavor of yogurt each day: blueberry (probability 0.2), banana (probability 0.5), and jalapẽno (probability 0.3). You have a mild banana allergy: every time you eat banana, you break out in a rash with probability 0.3. The other yogurts do not give you any adverse side effects.

(a) What is the probability you get a rash on a given day?

(b) Given that you do not get a rash on a given day, what is the probability the dining hall served banana yogurt that day?

(c) What is the expected number of times during one week (7 days) that you eat banana yogurt, but do not get a rash?

Solution:

(a) Let R be the event that you get a rash on a given day. Also let E_1, E_2, and E_3 be the events that you are served blueberry, banana, and jalapeno yogurt respectively on a given day. We are asked to calculate $\mathbb{P}(R)$. We can use the law of total probability to calculate this:

$$\mathbb{P}(R) = \sum_{i=1}^{3} \mathbb{P}(R|E_i)\mathbb{P}(E_i)$$

According to the given information we have:

$$\mathbb{P}(E_1) = 0.2,\ \mathbb{P}(E_2) = 0.5,\ \mathbb{P}(E_3) = 0.3,$$
$$\mathbb{P}(R|E_1) = 0,\ \mathbb{P}(R|E_2) = 0.3,\ \mathbb{P}(R|E_3) = 0$$

When we substitute the values, we get $\mathbb{P}(R) = 0.5 \cdot 0.3 = 0.15$.

(b) Let NR be the event that you don't get a rash on a given day. We are asked to calculate $\mathbb{P}(E_2|NR)$. By the Bayes' formula,

$$\mathbb{P}(E_2|NR) = \frac{\mathbb{P}(NR|E_2)\mathbb{P}(E_2)}{\mathbb{P}(NR)}$$

We know $\mathbb{P}(E_2) = 0.5$ and $\mathbb{P}(NR|E_2) = 1 - \mathbb{P}(R|E_2) = 0.7$. Since we already calculated $\mathbb{P}(R)$ in part a, we can directly say: $\mathbb{P}(NR) = 1 - \mathbb{P}(R) = 1 - 0.15 = 0.85$ since NR and R are complementary events.
Hence,

$$\mathbb{P}(E_2|NR) = \frac{0.7 \cdot 0.5}{0.85} = 0.412$$

(c) Let N be the number of times during one week that you eat banana yogurt, but do not get a rash. We are asked to find $\mathbb{E}[N]$.
Define N_i for i in $\{1, 2, \dots, 7\}$ as:

$$N_i = \begin{cases} 1 & \text{if you eat banana yogurt and do not get rash on day } i \\ 0 & \text{otherwise} \end{cases}$$

Note that we have: $N = \sum_{i=1}^{7} N_i$.

And by the linearity of expectation, we have: $\mathbb{E}[N] = \sum_{i=1}^{7} \mathbb{E}[N_i]$.

The N_i's are Bernoulli random variables with the same probability p of success. The parameter, p, of this random variable, is equal to the probability of eating a banana yogurt and not getting a rash on any given day.

So $p = \mathbb{P}(NR|E_2)\mathbb{P}(E_2) = 0.7 \cdot 0.5 = 0.35$, and $\mathbb{E}[N_i]$ is therefore equal to 0.35 as well.

Finally, $\mathbb{E}[N] = \sum_{i=1}^{7} 0.35 = 2.45$.

Midterm 1 2015 - Q3 (Rock, paper, scissors)

This classic two-player game is played as follows. There are three different hand gestures representing **r**ock, **p**aper, or **s**cissors. In every round of the game, each player simultaneously makes one of the gestures (so neither player knows what the other is going to choose). If both players make the same gesture, the round is a tie. Otherwise, one of the players wins according to the following rules: **r** wins against **s**, **p** wins against **r**, and **s** wins against **p**. When the players tie, they keep playing additional rounds until one of them wins, at which point the game ends.

As each player cannot know what the other will play, the best they can do is the following strategy: in each round, each player independently chooses one of the gestures **r**, **p**, **s** with equal probability (that is, each player and each round is independent).

(a) What is the probability that the first round is a tie?

(b) What is the expected number of rounds played until someone wins?

(c) What is the probability that the first player eventually wins the game?

(d) What is the expected number of times the first player plays r throughout the game?

Solution:

Let us denote by

$$(r, r), (r, p), (r, s), (p, r), (p, p), (p, s), (s, r), (s, p), (s, s)$$

the result of a pair of gestures, where the first letter denotes the gesture of player 1 and the second letter denotes the gesture of player 2. Let Xk be the outcome of the kth round of gestures. The random variable X_k takes one of the values in $\{(r, r), (r, p), (r, s), (p, r), (p, p), (p, s), (s, r), (s, p), (s, s)\}$.

(a) $\mathbb{P}(\text{First round is a tie}) = \mathbb{P}(X_1 \in \{(r, r), (p, p), (s, s)\}) = 3/9 = 1/3$

(b) For ease of notation let us define the following:

$$W := (r, s), (p, r), (s, p), (s, r), (r, p), (p, s)$$
$$T := (r, r), (p, r), (s, s)$$

Someone wins round k if $X_k \in W$. This happens with probability $\mathbb{P}(X_k \in W) = 6/9 = 2/3$. Round ends in a tie if $X_k \in T$ with probability $\mathbb{P}(X_k \in T) = 3/9 = 1/3$.

Let N denote the number of rounds until someone wins. We observe the following

- Each round can be seen as a Bernoulli trial where success is someone winning.
- Each round is independent from other rounds.
- The probability of success is 2/3. This probability, and therefore probability of failure as well remains constant from trial to trial.

We saw in class that a random variable that satisfies these conditions follows a Geometric distribution hence $N \sim \mathcal{G}(2/3)$. Thus the expected number of rounds played until someone wins is $\mathbb{E}(N) = 1/(2/3) = 3/2$. This comes from the fact that a Geometric(p) random variable has mean $1/p$.

(c) We observe that player 1 wins round k if and only if the first $k-1$ rounds are ties and the k-th round is a win for player 1. So

$$\{\text{Player 1 wins the k-th round}\} = \{X_1 \in T, \cdots, X_{k-1} \in T, X_K \in \{(r,s),(p,r),(s,p)\}\}$$

We take probabilities to get:

$$\begin{aligned}
\mathbb{P}(\text{Player 1 wins the kth round}) &= \mathbb{P}(X_1 \in T, \cdots, X_{k-1} \in T, X_K \in \{(r,s),(p,r),(s,p)\}) \\
&= \mathbb{P}\Big(\{X_1 \in T\} \cap \cdots \cap \{X_{k-1} \in T \cap \{X_K \in \{(r,s),(p,r),(s,p)\}\}\Big) \\
&= \mathbb{P}(X_1 \in T) \times \cdots \times \mathbb{P}(X_{k-1} \in T)\mathbb{P}(X_K \in \{(r,s),(p,r),(s,p)\}) \\
&\quad \text{by independence of the } X_k's \\
&= (\frac{1}{3})^{k-1}\frac{1}{3} \\
&= (\frac{1}{3})^k
\end{aligned}$$

The event “Player 1 eventually winning the gam” can be written as the infinite union of disjoint set as follows:

$$\bigcup_{k=1}^{\infty}\{\text{Player 1 wins round } k\}$$

We get:

$$\begin{aligned}
\mathbb{P}(\text{Player 1 wins eventually}) &= \mathbb{P}(\bigcup_{k=1}^{\infty}\{\text{Player 1 wins round } k\}) \\
&= \sum_{k=1}^{\infty}\mathbb{P}(\text{Player 1 wins round } k) \quad \text{since the events are disjoint} \\
&= \sum_{k=1}^{\infty}\left(\frac{1}{3}\right)^k \\
&= \frac{1/3}{1-1.3} = 1/2
\end{aligned}$$

(d) In each round, player 1 has a $1/3$ probability of choosing r and $2/3$ probability of doing otherwise. As before, let N be the number of rounds until someone wins.

Recall that $N \sim \mathcal{G}(2/3)$. Let R be the total number of r played by player 1. We define the following indicator random variable

$$U_k = \begin{cases} \text{1 if player 1 plays r in round k} \\ \text{0 otherwise} \end{cases}$$

We see that R can be written as a sum of indicator random variables U_k as follows:

$$R = \sum_{k=1}^{N} U_k$$

Suppose X_k is the outcome of the first player in round k and Y_k is the outcome of the second player in round k. We observe that

$$\begin{aligned} \mathbb{P}(X_k = r, X_k \neq Y_k) &= \mathbb{P}(X_k = r, Y_k = p \text{ or } s) \\ &= \mathbb{P}(X_k = r)\mathbb{P}(Y_k = p \text{ or } s) \text{ by independence} \\ &= \mathbb{P}(X_k = r) \times \frac{2}{3} \\ &= \mathbb{P}(X_k = r)\mathbb{P}(X_k \neq Y_k) \end{aligned}$$

which implies that N , the first time that $X_k \neq Y_k$, is independent of the events $\{X_k = r\}$.

The expected number of times the first player plays **r** throughout the game is

$$
\begin{aligned}
\mathbb{E}\Big[R\Big] &= \mathbb{E}\Big[\mathbb{E}\Big[R|N\Big]\Big] \text{ by the Tower property} \\
&= \sum_{n=1}^{\infty} \mathbb{E}(R|N=n)\mathbb{P}\{N=n\} \\
&= \sum_{n=1}^{\infty} \mathbb{E}\Big[\sum_{k=1}^{N} U_k|N=n\Big]\mathbb{P}(N=n) \\
&= \sum_{n=1}^{\infty} \mathbb{E}\Big[\sum_{k=1}^{n} U_k|N=n\Big]\mathbb{P}(N=n) \\
&= \sum_{n=1}^{\infty} \mathbb{E}\Big[\sum_{k=1}^{n} U_k\Big]\mathbb{P}(N=n) \text{ by independence of } U_k\text{s and N} \\
&= \sum_{n=1}^{\infty}\sum_{k=1}^{n} \mathbb{E}\Big[U_k\Big]\mathbb{P}(N=n) \quad \text{ by linearity} \\
&= \sum_{n=1}^{\infty} \Big(n \times \frac{1}{3}\Big)\mathbb{P}(N=n) \\
&= \frac{1}{3}\sum_{n=1}^{\infty} n\mathbb{P}(N=n) \\
&= \frac{1}{3}\mathbb{E}[N] \\
&= \frac{1}{3}\cdot\frac{3}{2} \quad \text{ from b} \\
&= \frac{1}{2}
\end{aligned}
$$

Midterm 1 2019 - Q1 (Questions on the exam) - *time permitting*

The exam has three questions. They are in increasing order of difficulty: you get the first question right with probability 3/4; the second question right with probability 1/2; and the third question right with probability 1/4. Your success in answering each question is independent.

(a) What is the expected number of questions you get right?

(b) What is the variance of the number of questions you get right?

(c) Given that you got two out of three questions right, what is the probability that you answered question number three correctly?

Solution:

(a) Let A_i be the event that you answer the i-th question correctly. Then the overall number Q of questions you answer can be written as $Q = \sum_{i=1}^{3} \mathbb{1}_{A_i}$. By linearity of the expectation,

$$\mathbb{E}[Q] = \mathbb{E}\Big[\sum_{i=1}^{3} \mathbb{1}_{A_i}\Big] = \sum_{i=1}^{3} \mathbb{E}[\mathbb{1}_{A_i}] = \sum_{i=1}^{3} \mathbb{P}(A_i) = \frac{3}{4} + \frac{1}{2} + \frac{1}{4} = \frac{3}{2}$$

(b) By the independence of the $\mathbb{1}_{A_i}$'s, the variance can be calculated as:

$$\begin{aligned}\operatorname{Var}[Q] = \operatorname{Var}\left[\sum_{i=1}^{3}\mathbb{1}_{A_i}\right] = \sum_{i=1}^{3}\operatorname{Var}\left[\mathbb{1}_{A_i}\right] = \sum_{i=1}^{3}\mathbb{P}(A_i)(1-\mathbb{P}(A_i))\\ = \frac{3}{4}\times\frac{1}{4}+\frac{1}{2}\times\frac{1}{2}+\frac{1}{4}\times\frac{3}{4} = \frac{5}{8}\end{aligned}$$

(c) We are looking to calculte $\mathbb{P}(A_3|Q=2)$. By definition of the conditional probability, we have:

$$\begin{aligned}\mathbb{P}(A_3|Q=2) &= \frac{\mathbb{P}(A_3, Q=2)}{\mathbb{P}(Q=2)}\\ &= \frac{\mathbb{P}(A_1\cap A_2^c\cap A_3)+\mathbb{P}(A_1^c\cap A_2\cap A_3)}{\mathbb{P}(Q=2)}\end{aligned}$$

Since $\{Q=2\} = \{A_1^c\cap A_2\cap A_3\}\cup\{A_1\cap A_2^c\cap A_3\}\cup\{A_1\cap A_2\cap A_3^c\}$ and by independence of the $\mathbb{1}_{A_i}$, we have:

$$\begin{aligned}\mathbb{P}(Q=2) = \mathbb{P}(A_1^C)\mathbb{P}(A_2)\mathbb{P}(A_3)+\mathbb{P}(A_1^C)\mathbb{P}(A_2)\mathbb{P}(A_3)+\mathbb{P}(A_1)\mathbb{P}(A_2)\mathbb{P}(A_3^C)\\ = \frac{3}{4}\times\frac{1}{2}\times\frac{3}{4}+\frac{3}{4}\times\frac{1}{2}\times\frac{3}{4}+\frac{1}{4}\times\frac{1}{4}\times\frac{1}{4} = \frac{13}{32}\end{aligned}$$

and by a similar method, we can also calculate the numerator for $\mathbb{P}(A_3|Q=2)$ to obtain:

$$\begin{aligned}\mathbb{P}(A_3|Q=2) &= \frac{\frac{3}{4}\times\frac{1}{2}\times\frac{3}{4}+\frac{3}{4}\times\frac{1}{2}\times\frac{3}{4}}{\frac{13}{32}}\\ &= \frac{4}{13}\end{aligned}$$